Universal Phonetic Embeddings for Cross-Script Toponym Matching: A Teacher-Student Approach

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Abstract

Linking place names across historical sources, languages, and writing systems remains a fundamental challenge in digital humanities and geographic information retrieval. Existing approaches either require language-specific phonetic algorithms, expert-curated transliteration rules, or fail to capture the phonetic relationships that make "Moscow", "Москва", and "موسكو" recognisably the same place. We present a neural embedding system that maps toponyms from any script into a unified 128-dimensional phonetic space, enabling direct similarity comparison without runtime phonetic conversion or language pair-specific resources. Our approach uses a Teacher-Student architecture: a Teacher network trained on articulatory phonetic features (derived via Epitran grapheme-to-phoneme conversion and PanPhon feature extraction) produces target embeddings, while a Student network learns to approximate these embeddings directly from characters, guided by script and optional language embeddings. The Student operates at inference time without requiring phonetic resources, enabling deployment at scale. We train on co-located toponym pairs from GeoNames, Wikidata, and TGN, applying phonetic similarity filtering to exclude false cognates. A three-phase curriculum progresses from phonetic feature learning through knowledge distillation to hard negative discrimination. [Results pending: cross-script retrieval accuracy, noise robustness.]

1 Introduction

Historical gazetteers aggregate place references from sources spanning millennia, dozens of languages, and multiple writing systems. The World Historical Gazetteer (WHG) alone indexes nearly [67] million toponym records from antiquity to the present, drawn from sources as diverse as classical Latin itineraries, medieval Arabic geographies, and modern national mapping agencies. These records are distinct but include language tags, so there is naturally some replication of unlanguaged name forms across records. Linking these references—determining that "Londinium", "Londres", "Лондон", and "伦敦" all denote the same city—is essential for cross-cultural historical research but remains technically challenging.

The core difficulty is that place name similarity is fundamentally *phonetic*, not orthographic. English speakers recognise "Munich" and "München" as the same city not because the spellings match, but because the sounds are similar enough to suggest common origin. Yet computational approaches to toponym matching have largely relied on string-based metrics (edit distance, Jaro-Winkler) or phonetic algorithms designed for single languages (Soundex, Metaphone for English; Cologne phonetic for German). These approaches fail catastrophically when names cross script boundaries: no amount of edit distance tuning will link "東京" to "Tokyo".

Recent work has applied neural embeddings to geographic entity matching, but existing systems either operate within single scripts, require language identification at inference time, or depend on curated transliteration resources that don't scale to the long tail of historical languages. Until now, we have lacked a system that can place "Москва", "Moscow", "موسكو", and "モスクワ" near each other in embedding space using only the raw character sequences as input.

This paper presents such a system. Our key contributions are:

1. A **script-aware but script-agnostic embedding architecture** that maps toponyms from 20+ writing systems into a unified 128-dimensional space where proximity reflects phonetic similarity.

2. A **Teacher-Student training framework** that transfers phonetic knowledge from articulatory features (available only for IPA-transcribed names) to a character-level model requiring no phonetic resources at inference time.

3. A **noise-aware training regime** with strategic language dropout that forces the model to learn script-intrinsic phonetic patterns rather than relying on language-specific shortcuts.

4. **Phonetic similarity filtering** for training data that prevents the model from learning false equivalences between unrelated exonyms (e.g., "Germany" $\neq$ "Deutschland").

5. A **three-phase curriculum** progressing from phonetic feature learning through knowledge distillation to hard negative discrimination.

The resulting system enables fuzzy phonetic search across the full WHG corpus without language identification, script-specific preprocessing, or runtime phonetic conversion.

Scope and Integration. This work addresses the *name-matching* component of toponym resolution specifically. The model has no access to geographic coordinates or spatial context—it operates purely on phonetic similarity between strings.

In WHG's reconciliation pipeline (termed WHG PLACE: Place Linkage, Alignment, and Concordance Engine), phonetic similarity serves as one input to a larger process: candidate toponyms are first retrieved via the phonetic-aware toponym index, then places containing matching toponyms are selected from the places index and filtered by geographic proximity and other contextual criteria. The contribution here is the phonetic similarity component alone.

2 Related Work

2.1 Classical Phonetic Algorithms

The earliest computational approaches to phonetic matching emerged from English-language record linkage. Soundex (Russell, 1918), developed for US Census indexing, maps names to four-character codes by retaining the first letter and encoding subsequent consonants. Its English-centricity is explicit: the algorithm assumes Latin script and English phoneme-grapheme correspondences. Metaphone (Philips, 1990) and Double Metaphone (Philips, 2000) improve accuracy for English by handling common spelling variations, but remain fundamentally tied to English phonology.

Language-specific variants exist for other European languages, but each requires separate implementation and fails outside its target language family. No classical algorithm handles cross-script matching: they assume a single alphabet and a known phoneme inventory.

2.2 String Similarity Metrics

Edit distance metrics (Levenshtein, Damerau-Levenshtein, Jaro-Winkler) measure orthographic rather than phonetic similarity. While useful for detecting OCR errors or spelling variants within a script, they cannot capture phonetic relationships across scripts. The Levenshtein distance between "Moscow" and "Москва" is undefined without transliteration; even after romanisation to "Moskva", the distance of 2 fails to reflect the near-identical pronunciation.

Recchia and Louwerse (Recchia and Louwerse, 2013) evaluated 21 similarity measures on romanised toponyms from 11 countries, finding that optimal measures varied by language but were consistent within linguistic families. Santos et al. (Santos et al., 2018) demonstrated that combining multiple string metrics with machine learning classifiers improved matching accuracy for Portuguese toponyms.

Phonetic-aware edit distances (Kondrak, 2000) weight substitutions by articulatory similarity, but still require input in a common alphabet.

2.3 Neural Approaches to Entity Matching

Recent work applies neural embeddings to geographic entity resolution. Qiu et al. (Qiu et al., 2024) proposed GeoBERT, combining a BERT model pre-trained on Chinese geographic text with the Enhanced Sequential Inference Model (ESIM) architecture for toponym pair classification. Their approach integrates multiple Chinese-specific features including Pinyin romanisation, Wubi keyboard encoding, character radicals, and stroke counts. On Chinese address matching datasets, GeoBERT achieved F1 scores exceeding 0.97. However, the Chinese-specific features have no analogues in other writing systems, and the geographic pre-training corpus is monolingual, limiting cross-lingual application.

Most relevant is the MEHDIE project (Sagi et al., 2025), which developed tools for matching toponyms between historical Hebrew and Arabic sources. Sagi et al. present a benchmark comprising place pairs from 13th-century Arabic geographical texts and 12th-century Hebrew travel narratives. Their approach investigates direct transliteration between Hebrew and Arabic scripts using a custom rule-based library that exploits the phonetic affinity between Semitic languages. They find that direct transliteration between related scripts often outperforms romanisation due to the loss of phonetic information in standard Latin transliteration schemes. The MEHDIE work validates our central hypothesis—that phonetic relationships between languages can be exploited for toponym matching—while highlighting the limitations of language-pair-specific approaches.

Rama (Rama, 2016) applied Siamese convolutional networks to cognate identification using character embeddings, demonstrating that neural architectures can learn sound correspondences between related languages. Our work differs in using explicit phonetic features via PanPhon rather than learning phonetic patterns implicitly from characters alone.

2.4 Cross-Lingual Representation Learning

The broader literature on cross-lingual embeddings provides relevant techniques. Conneau et al. (Conneau et al., 2018) show that monolingual embedding spaces can be aligned post-hoc using small bilingual dictionaries. Artetxe et al. (Artetxe et al., 2018) demonstrate unsupervised alignment using only monolingual corpora. However, these approaches target semantic similarity ("dog" near "chien") rather than phonetic similarity ("Paris" near "Париж").

The TRANSLIT resource (Benites et al., 2020) provides 1.6 million name variants across 180 languages for training transliteration systems. While valuable, such resources do not directly address the phonetic similarity task targeted here.

2.5 Phonetic Representations in NLP

The use of phonetic features in natural language processing has a long history. Epitran (Mortensen et al., 2018) provides rule-based grapheme-to-phoneme conversion for over 100 languages, outputting IPA transcriptions. PanPhon (Mortensen et al., 2016) maps IPA segments to articulatory feature vectors encoding phonological properties. These resources form the foundation of our Teacher network's phonetic grounding.

2.6 Siamese Networks for Similarity Learning

Siamese networks (Bromley et al., 1993; Chopra et al., 2005) learn similarity functions by processing two inputs through identical sub-networks and comparing their output representations. They have been successfully applied to face verification (Schroff et al., 2015), signature verification, and text similarity (Mueller and Thyagarajan, 2016; Neculoiu et al., 2016).

For text matching, Mueller and Thyagarajan (Mueller and Thyagarajan, 2016) demonstrated that Siamese LSTMs with Manhattan distance achieve strong performance on sentence similarity tasks. Lin et al. (Lin et al., 2017) introduced structured self-attention for sentence embeddings. Our architecture extends this line of work by incorporating self-attention mechanisms and operating on phonetic feature sequences.

3 Architecture

Our system employs a Teacher-Student architecture where a Teacher network, trained on articulatory phonetic features, produces target embeddings that a Student network learns to approximate from character sequences alone. At inference time, only the Student is required, enabling deployment without phonetic resources.

3.1 Design Principles

Three principles guide our architecture:

Script Awareness Without Script Dependence. The model must handle 20 writing systems but produce embeddings in a unified space where script boundaries are transparent. We achieve this through deterministic script detection (via Unicode block analysis) combined with learned script embeddings that allow the model to interpret characters appropriately for each script.

Phonetic Grounding. Embedding similarity must reflect phonetic similarity, not orthographic or semantic similarity. We ground the embedding space through the Teacher network, which operates on articulatory features derived from IPA transcriptions.

Inference Efficiency. The deployed model must encode toponyms without runtime phonetic conversion, language identification, or external resources. All phonetic knowledge is distilled into the Student's parameters during training.

3.2 Script Detection and Character Vocabulary

We detect script from Unicode code points, mapping each character to one of 20 script categories: Latin, Cyrillic, Greek, Arabic, Hebrew, Devanagari, Bengali, Tamil, Telugu, Malayalam, Kannada, Gujarati, Thai, Georgian, Armenian, Chinese (Han), Japanese (Hiragana, Katakana), Korean (Hangul), and Other.

Character vocabulary construction balances coverage against tractability:

- **Alphabetic scripts** (Latin, Cyrillic, Greek, Arabic, Hebrew, Brahmic scripts, Georgian, Armenian, Thai): We retain native characters, yielding vocabularies of 50–200 tokens per script.
- **Chinese and Japanese**: The tens of thousands of Han characters would create an intractable vocabulary. We romanise via AnyAscii, reducing to the Latin character set while preserving approximate phonetic content.
- **Korean Hangul**: Rather than treating 11,172 syllable blocks as atomic units, we decompose into Jamo (lead consonant, vowel, optional tail consonant), yielding approximately 60 tokens with explicit phonetic structure.

The resulting vocabulary contains 3,634 tokens across all scripts (and 829 languages), each tagged with its source script for embedding lookup.

3.3 Teacher Network: Phonetic Feature Encoder

The Teacher network encodes toponyms via their IPA transcriptions and articulatory features. Given a toponym, we first obtain an IPA transcription using Epitran (Mortensen et al., 2018), then extract PanPhon features (Mortensen et al., 2016)—a 24-dimensional binary vector per phoneme encoding articulatory properties (place, manner, voicing, etc.).

The Teacher architecture:

1. **Input**: Sequence of PanPhon feature vectors, $(f_1, f_2, \ldots, f_T)$ where $f_i \in \{0, 1\}^{24}$
2. **Projection**: Linear layer projects features to hidden dimension: $h_i = W_f f_i + b_f$
3. **BiLSTM**: Bidirectional LSTM processes the sequence: $(\overrightarrow{h_i}, \overleftarrow{h_i}) = \text{BiLSTM}(h_1, \ldots, h_T)$
4. **Self-Attention**: Multi-head self-attention captures long-range dependencies
5. **Attention Pooling**: Learned attention weights aggregate sequence to fixed-length vector
6. **Output Projection**: Linear layer to 128 dimensions with L2 normalisation

The Teacher is trained on triplets of co-located toponyms, learning to place phonetically similar names (same place, different languages) closer than phonetically dissimilar names (different places).

3.4 Student Network: Universal Character Encoder

The Student network processes raw character sequences, optionally augmented with script and language information:

1. **Character Embedding**: Each character maps to a learned 96-dimensional vector via script-partitioned vocabulary lookup
2. **Script Embedding**: Deterministically detected script maps to a learned 16-dimensional vector, concatenated to each character embedding
3. **Language Embedding**: When available, ISO 639 language code maps to a learned 16-dimensional vector; otherwise, a special `<UNK>` token is used. During training, known languages are randomly replaced with `<UNK>` with 50% probability, forcing the model to learn script-intrinsic patterns.
4. **BiLSTM + Self-Attention + Attention Pooling**: Identical architecture to Teacher
5. **Output Projection**: Linear layer to 128 dimensions with L2 normalisation

3.5 Noise Augmentation

To handle OCR errors, historical spelling variations, and transcription inconsistencies, we apply character-level noise during Student training:

- **Insertion**: Random character from same script inserted (probability 0.1)
- **Deletion**: Random character deleted (probability 0.1)
- **Substitution**: Random character replaced with another from same script (probability 0.05)
- **Transposition**: Adjacent characters swapped (probability 0.05)

Noise is applied with 30% probability per training example, with the target being the Teacher's embedding of the *clean* input. This trains the Student to map corrupted inputs to the same phonetic region as their clean counterparts.

4 Training Data

4.1 Data Selection and Curation Criteria

The World Historical Gazetteer (WHG) infrastructure separates data into a `places` index, which aggregates geographic entities with their attributions, and a `toponyms` index, which abstracts individual name variants by script and language. Our data selection strategy leverages this architecture to filter high-quality training data from major namespace authorities—GeoNames (`gn`), Wikidata (`wd`), and the Getty Thesaurus of Geographic Names (`tgn`)—while excluding less strictly curated sources like OpenStreetMap.

Training requires pairs of toponyms that refer to the same place. We extract these from co-located name attestations: if an authority lists "London", "Londres", "Londra", and "Лондон" within the same 'place', we generate pairs among all combinations.

To address the inherent script and language imbalances in the raw corpus, we define our training data selection based on five strict criteria.

1. Script-Stratified Sampling. Rather than ingesting all available pairs from the `toponyms` index, which would result in a corpus heavily skewed toward Latin script and European languages, we implement a quota-based sampling strategy. We define target script pairs (e.g., Latin-Cyrillic, Latin-Devanagari, Arabic-Hebrew) and sample up to $N = 100,000$ pairs for each combination. This prevents high-resource scripts from dominating the gradient descent while ensuring sufficient signal for low-resource script pairs.

2. Namespace Filtering and Coverage. We restrict training data to records from the `gn`, `tgn`, and `wd` namespaces. Within this allowed pool, the inclusion of `wd` (Wikidata) is critical for meeting the stratification quotas defined above. As a direct result of our sampling criteria, Wikidata records are frequently selected to populate South Asian (Devanagari, Tamil, Telugu, Kannada, Malayalam, Bengali) and Southeast Asian (Thai, Khmer, Lao, Myanmar) script pairs, where other gazetteers lack the necessary depth of multi-script correspondence.

3. Global Vocabulary Construction. To prevent out-of-vocabulary (OOV) errors at inference time, we decouple vocabulary construction from pair generation.

- **Pass 1 (Vocabulary):** We scan the entire `toponyms` index (representing 67M records) to build the character vocabulary. Crucially, this scan is followed by script-specific preprocessing: CJK and Japanese Kana are romanised via AnyAscii to reduce the intractable logographic space to Latin characters , while Korean Hangul is decomposed into constitutive Jamo to expose phonetic structure. This ensures the vocabulary captures every token the model will encounter, covering rare characters in scripts retained in their native forms (e.g., Cyrillic, Arabic, Devanagari) while handling large-inventory scripts efficiently.

- **Pass 2 (Training Pairs):** We generate candidate triplets from the `places` index, which groups name variants by entity. These candidates are then restricted to the stratified quotas defined above.

4. Isorthographic Phonetic Consistency. We explicitly retain "same-string, different-language" pairs (isorthographs) regardless of whether their pronunciation differs.

- **Teacher Signal:** The Teacher network receives sequences of PanPhon articulatory feature vectors derived from IPA transcriptions. It provides the ground truth: if the pronunciation differs (e.g., "Paris" in French vs. English), the Teacher pushes the target embeddings apart; if pronunciation is stable (e.g., "Berlin" in German vs. English), it keeps them close.

- **Student Signal:** The Student receives identical character sequences but distinct Language Embeddings.

By retaining both diverging and stable isorthographs, we force the Student network to learn when to rely on the Language Embedding to modify its phonetic prediction and when the script alone is sufficient.

5. Cross-Script Phonetic Alignment. We prioritise pairs that maximize script distance while minimizing phonetic distance. During sampling from the `places` index, we heavily weight pairs where the source and target scripts differ (e.g., 北京 vs. Beijing) over same-script variants (Beijing vs. Peking), as the primary architectural goal is to bridge disjoint character spaces.

4.2 Dataset Composition

Following these criteria, the training distribution is forcibly balanced. While the total volume of Latin script tokens remains high due to its ubiquity as a pivot script, the relative representation of Devanagari, Bengali, Tamil, Telugu, Thai, and other under-resourced scripts is boosted to meet minimum training thresholds.

4.3 Pair Generation with Phonetic Filtering

A critical challenge is **false cognates**: exonyms that refer to the same place but share no phonetic relationship. For example, "Germany" and "Deutschland", or "日本" (Nihon) and "Japan", denote the same entity but should not be trained as phonetically similar. Naively including such pairs would corrupt the embedding space.

We apply phonetic similarity filtering using a simple heuristic: we compute normalised Levenshtein similarity between name pairs (after romanisation via AnyAscii in the case of CJK). Pairs below a threshold of 0.35 are excluded from positive training examples. This successfully filters:

- "Germany" / "Deutschland" (similarity 0.18)
- "Japan" / "日本" $\rightarrow$ "Ri Ben" (similarity 0.22)
- "Finland" / "Suomi" (similarity 0.15)

While retaining genuine phonetic cognates:

- “London” / “Londres” (similarity 0.57)
- “München” / “Munich” (similarity 0.57)
- “Москва” $\rightarrow$ “Moskva” / “Moscow” (similarity 0.71)

4.4 IPA Transcription

The Teacher network requires IPA transcriptions. We use Epitran (Mortensen et al., 2018) for grapheme-to-phoneme conversion, which supports approximately 100 languages covering the majority of GeoNames entries. Epitran provides rule-based G2P that outputs IPA transcriptions suitable for PanPhon feature extraction. Names in languages without Epitran support are excluded from Teacher training but can still be processed by the Student, which learns to generalise from related scripts.

4.5 Dataset Statistics

[REVISE]

Training data comprises 17.1 million toponym records drawn from three sources: GeoNames (17M, 99.8%), Pleiades (24,768), and Index Villaris (16,995). Of these, 4.4 million (26%) have phonetic embeddings via Epitran and PanPhon suitable for Teacher training. The data spans 829 languages and 20 script categories.

Positive pairs are generated dynamically during training from co-located name attestations sharing geographic coordinates. After deduplication, 2.55 million unique pairs remain, of which 45.3% are cross-script. Analysis of pair frequency revealed significant redundancy from common place names (Georgetown, Springfield, Salem) appearing across multiple locations.

- Total toponym tokens: 17.1 million
- Script distribution (top 10 identified): Latin (14.2M, 83.3%), Cyrillic (816k, 4.8%), CJK (752k, 4.4%), Arabic (709k, 4.2%), Thai (166k, 1.0%), Hangul (144k, 0.8%), Other (111k, 0.6%), Katakana (64k, 0.4%), Hiragana (41k, 0.2%), Armenian (38k, 0.2%).
- Language distribution (Top 20 identified): English (647k), Chinese (604k), Norwegian (427k), Russian (398k), Spanish (349k), Persian (307k), Indonesian (282k), Finnish (241k), Arabic (231k), French (179k), Thai (176k), Japanese (171k), Portuguese (147k), Ukrainian (117k), German (117k), Korean (113k), Dutch (74k), Latin-Uncertain/Other (72k), Italian (70k), Swedish (68k).

5 Training Methodology

We employ a three-phase curriculum that progressively builds from phonetic feature learning through knowledge distillation to discriminative fine-tuning.

5.1 Phase 1: Teacher Training on Phonetic Features

The Teacher network learns to produce embeddings where phonetically similar toponyms cluster together. Training uses triplet margin loss:

$$\mathcal{L}_{\text{triplet}} = \max(0, \|e_a - e_p\|_2 - \|e_a - e_n\|_2 + m) \tag{1}$$

where e_a, e_p, e_n are embeddings of anchor, positive (same place), and negative (different place) toponyms, and m is the margin (we use $m = 0.3$).

Negatives are sampled from the same batch (in-batch negatives), ensuring the Teacher learns to distinguish phonetically similar but geographically distinct names.

Training Configuration.

- Optimiser: AdamW with weight decay 0.01
- Learning rate: 5×10^{-4} with cosine annealing
- Batch size: 256 triplets
- Epochs: 50

5.2 Phase 2: Student-Teacher Alignment

The Student learns to approximate Teacher embeddings from character sequences. Given a toponym with both character sequence (Student input) and IPA transcription (Teacher input) [Shouldn't this be PanPhon embedding?], we minimise the distance between their embeddings:

$$\mathcal{L}_{\text{distill}} = \alpha \cdot \text{MSE}(e_S, e_T) + (1 - \alpha) \cdot (1 - \cos(e_S, e_T)) \tag{2}$$

where e_S and e_T are Student and Teacher embeddings, and $\alpha = 0.5$ balances MSE and cosine components.

During this phase:

- Language dropout (50%) forces learning of script-intrinsic patterns
- Noise augmentation (30%) trains robustness to input corruption
- Teacher weights are frozen

Training Configuration.

- Optimiser: AdamW with weight decay 0.01
- Learning rate: 3×10^{-4} with cosine annealing
- Batch size: 512
- Epochs: 50

5.3 Stage 3: Discriminative Fine-Tuning

While the distillation phase aligns the Student with the Teacher's general phonetic space, the Student may still struggle with *minimal pairs*—names with high orthographic similarity but distinct phonetic realizations (e.g., "Austria" vs. "Australia", or "Laughter" vs. "Slaughter").

In this final phase, we fine-tune the Student using triplet loss to sharpen these boundaries without degrading the learned cross-script equivalences.

Hard Negative Mining Strategy. We define negatives strictly phonetically by mining triplets (a, p, n) where:

- **Anchor (a) & Positive (p):** A cross-script pair confirmed to be phonetically equivalent by the Teacher (low Teacher-space distance).
- **Hard Negative (n):** A toponym selected such that it has high orthographic similarity to the anchor (Levenshtein similarity > 0.7) but is phonetically distinct (high Teacher-space distance).

Crucially, we explicitly **exclude** isorthographs from being selected as negatives. If two records share the name "Georgetown" but refer to different locations, they are treated as identical in the embedding space. The loss function penalizes the Student only when it places phonetically distinct names closer than phonetically equivalent ones.

Training Configuration.

- **Loss Function:** Triplet Margin Loss ($m = 0.2$).
- **Teacher Status:** Frozen (used only to validate true phonetic distances for negative sampling).
- **Optimiser:** AdamW with a reduced learning rate (1×10^{-5}) to prevent catastrophic forgetting of the cross-script alignment learned in Stage 2.

5.4 Implementation Details

Training uses PyTorch on NVIDIA A100 GPUs. The Teacher contains approximately 2M parameters; the Student approximately 4M parameters (larger due to character vocabulary). Phase 1 completes in approximately 2 hours; Phase 2 in approximately 9 hours; Phase 3 in approximately 12 hours.

Checkpoints are saved after each epoch, with the best model selected by validation loss.

6 Evaluation

6.1 Preliminary Results

[DEPRECATED: THESE RESULTS ARE FROM A PROTOTYPE MODEL]

We present interim results from initial training runs to characterise model behaviour prior to full evaluation. These results use the Phase 2 checkpoint (Student-Teacher alignment complete, before hard negative fine-tuning).

Cross-script equivalents. As shown in Table 1, these pairs achieve high similarity as intended.

Name 1	Name 2	Similarity
Moskva	Москва	0.998
Parizh	Париж	0.995
Moscow	Moscou	0.986
London	Лондон	0.997

Table 1: Cross-script transliteration pairs.

Historical variants. These pairs show moderate to strong similarity, with results summarised in Table 2.

Modern	Historical	Similarity
London	Londinium	0.938
Cologne	Colonia	0.968
Lyon	Lugdunum	0.683
Istanbul	Constantinople	0.582

Table 2: Historical variant pairs.

Unrelated pairs. Conversely, unrelated pairs are appropriately separated (Table 3).

Name 1	Name 2	Similarity
London	Tokyo	0.335
Paris	Beijing	0.268
Berlin	Cairo	-0.116

Table 3: Unrelated toponym pairs.

The pairwise similarity distribution across 22 diverse toponyms shows mean 0.092 ($\sigma = 0.348$), indicating the embedding space has not collapsed and maintains discrimination between unrelated names.

Limitations observed. Same-script cross-lingual variants show weaker performance than cross-script pairs: London–Londres (0.754) and Munich–München (0.923) fall below the 0.95 target, suggesting the model relies partly on script differences as a cue for phonetic equivalence. Phase 3 hard negative training aims to address this but requires careful curation to avoid introducing false equivalences.

6.2 Evaluation Tasks

[This section describes the evaluation methodology. Results are pending completion of Phase 3 training.]

We evaluate on four tasks designed to test the core claims of the system:

Task 1: Cross-Script Retrieval. Given a toponym in one script, retrieve its phonetic equivalents in other scripts. We sample places from Wikidata that have labels (`skos:altLabel`) in at least 3 distinct scripts. For each place, we use one script variant as query and measure retrieval of the other script variants. Importantly, Wikidata was not used in training, providing an independent test set.

Metrics: Recall@1, Recall@5, Recall@10, Mean Reciprocal Rank (MRR).

Task 2: Noise Robustness. Measure embedding stability under synthetic noise simulating OCR errors and transcription mistakes. For each test toponym, we generate corrupted variants at increasing noise levels (5%, 10%, 20% character error rate) and measure cosine similarity between clean and corrupted embeddings.

Metrics: Mean cosine similarity at each noise level; retrieval accuracy for corrupted queries.

Task 3: Historical Variant Linking. Link historical spelling variants to their attested modern or period-appropriate forms. We use two approaches:

- Index Villaris 17th-century spellings linked to place names from Ordnance Survey maps c.1900 (the GB1900 project) and, where available, to Wikidata entries.
- Wikidata's historical name properties (e.g., `P1705` native label, various "historical name" properties) which may provide attestations independent of our training sources.

A key methodological concern is that training on GeoNames, Pleiades, and Index Villaris means evaluation on these same sources risks overfitting. We therefore prioritise evaluation on held-out Wikidata attestations not seen during training.

Metrics: Linking accuracy (correct form in top-k).

Task 4: Scalability. Measure retrieval performance over the full WHG corpus of 67M toponym records using Elasticsearch's approximate k-NN with HNSW indexing.

Metrics: Query latency (p50, p95, p99); recall at various index configurations.

6.3 Baselines

We compare against:

- **Levenshtein on AnyAscii**: Edit distance after romanisation
- **Jaro-Winkler on AnyAscii**: String similarity after romanisation
- **Double Metaphone**: Phonetic codes after romanisation
- **Elasticsearch fuzzy match**: Built-in fuzzy query with fuzziness=AUTO
- **mBERT cosine similarity**: Multilingual BERT [CLS] embeddings

6.4 Results

[Results tables and analysis pending.]

7 Discussion

[Discussion pending results.]

7.1 Limitations

Several limitations warrant acknowledgment:

Training Data Bias. Our training sources over-represent European languages and the Latin script. Performance on under-represented scripts (e.g., Ethiopic, Myanmar, Tibetan) may be weaker.

Tonal Languages. The current architecture does not explicitly model tone, which is phonemically contrastive in Chinese, Vietnamese, Thai, and other languages. Tonal minimal pairs may not be distinguished.

IPA Coverage. The Teacher's phonetic grounding depends on eSpeak-ng's G2P coverage. Languages without eSpeak support contribute only through the Student's character-level learning.

Computational Cost. Embedding 60M toponyms requires significant compute. While inference is fast (thousands of embeddings per second on GPU), full corpus re-embedding after model updates takes several hours.

8 Conclusion

We have presented a neural embedding system for cross-script toponym matching that places phonetically similar names near each other in a unified 128-dimensional space, regardless of writing system. The Teacher-Student architecture enables training on articulatory phonetic features while requiring only character sequences at inference time. Phonetic similarity filtering prevents learning of false cognates, and noise-aware training with language dropout forces the model to learn script-intrinsic phonetic patterns.

[Summary of results and impact pending.]

The system is deployed in the World Historical Gazetteer, enabling fuzzy phonetic search across 60M+ toponyms from antiquity to the present. We release our trained models and training code to support further research in historical toponym linking.

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[Additional acknowledgments pending.]

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